# TASKS OF III.RD INTERNATIONAL BIOLOGY OLYMPIAD

Practical Part

Dear Competitors,

In the practical part of the IBO you will examine and compare organisms living in water ecosystem (in a fishpond) and organisms living in terrestrial environment near the water. The practical part is divided into three distinct groups of tasks:

1st group: Tasks No. 1-7 (time app. 120 min.)
2nd group: Tasks No. 8-12 (time app. 120 min.)
3rd group: Tasks No. 13-16 (time app. 60 min.).

You will receive a separate set of answer sheets for each of the tasks groups. After the finishing each of the group you must give the sheets to the assistan. Then you can do an other group of tasks.

Living, preserved organisms or their parts, used for this examination, are marked by capital letters of the Latin alphabet in the same way in all tasks, figures, vesells with organisms as well as in the answer sheets.Used these instead of the scientific names of the organisms in all three groups of tasks.

To make the evaluation of your results easier, use Tables I-IV in labeling of your drawings. The tables contain lists of morphological and/or anatomical terms with number codes. Use these codes on your answer sheets, according to the requirements of the tasks.

## 1ST GROUP OF TASKS (two hours)

Number 1-7

During the next two hours you will examine of anatomical structure of some water organisms (A,B,G), one vascular plant from a fishpond shore (D) and two terrestrial organisms (E,T).

### Task No. 1: Examination of the organims B

Sheet P1, Table I, Score:

Prepare a microscopic slide. You receive a drop of water with organism B. Examine it by microscope.

Choose one cell and draw it (including inner structure) on the answer sheet P1.

Make a description of the cell parts in your drawing by number codes from Table I.

Task No. 2: Examination of the organism A
Sheet P1, Table I, Score:

This task is similar as task No. 1 but you will examine other species. Examine, draw and make a description as in task No. 1.

Task No. 3: Anatomical structure of a leaf of a vascular plant D
Sheet P2, Table I, Score:

There is a leaf of a plant D on the table. Made a cross-section of the leaf.

a/ Draw schematically distribution of tissues in the sheet P2.
b/ Draw a detail of a tissue which is characteristic (typical) for the plant species.

Task No. 4: Examination of the organism G
Sheet P3, Table III, Score:

Organisms G represents one of the most characteristic organisms living in a fishpond.

Transfer one animal with drop of water by a wide pipette on a slide with a pitlet/socket

Examine mainly the organs of locomotion and digestive system.

a/ Draw, into the scheme of the animal body in the sheet P3, digestive system, heart and antennulla position.
b/ Make a description by number codes from table III.
c/ Mark by arrows the organs of locomotion in the sheet P3.

Task No. 5: Examination of a leaf of a forest herb E
Sheet P4, Table I, Score:

Leaf of the plant E is in a vessel on the table.

Prepare a fresh microscopical sample of the epidermis from lower side of the leaf.

Draw in the sheet P4 a stoma and epidermal cells.

Make a description of the drawing structures by number codes from Table I.

Task No. 6: Anatomical structure of a stem of the pla.
Sheet P4, Table I, Score:

In a vessel you have a flowering shoot of the plant E.
Made a cross-section of the stem (better from the below of the flowers) and prepare fresh microscopic slide.

a/ Draw schematically a position of tissues in the sheet P4.
b/ Make a description of the figure by number codes from Table I.

Task No. 7: Examination of the organism T
Sheet P5, Table II, Score:

There is a native temporary microscopic slide with organims T on your table.
Examine marginal parts of the slide by microscope.

a/ Draw typical anatomy of the body of the organisms in the sheet P5.
b/ Make a description of your drawing by number codes from Table II.

If you finished of the 1st group of tasks, you control again your answers before colecting of the ans[illegible]eets P1-P5.

P1



Sheet P 1
лист

Frage Organismus
Task No.1 : Organism B
Задача № 1 : Организм B

Competitor No.
участник № . . . . .

Frage Organismus
Task No.2 : Organism A
Задача № 2 : Организм A

Sheet P 2
лист

P2

Competitor No. ......
участник №

-6-

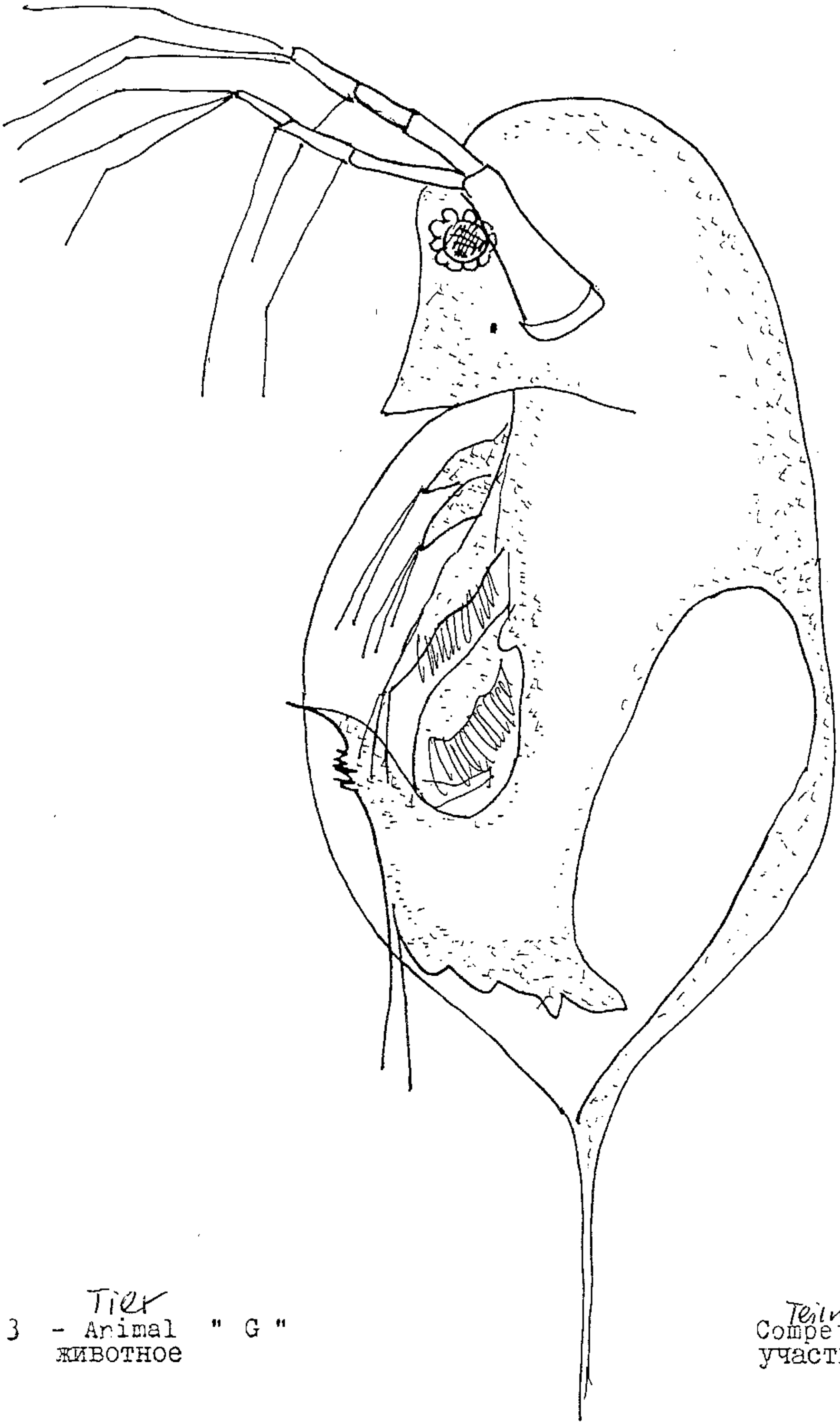
Task No.3 : Anatomical structure of leaf of the plant D
Задача №.3 : Анатомические структуры листа растения D

d. m.

Anatomische Struktur des Blattes der Pflanze D

P3





Tier
P 3 - Animal " G "
животное

Teilnehmer
Competitor No.:
участник №.

# TASKS OF III.RD INTERNATIONAL BIOLOGY OLYMPIAD

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To make the evaluation of your results easier, you will use Tables I-IV in labeling your drawings. The tables contain lists of morphological and/or anatomical terms with number codes. Use these codes on your answer sheets, according to the requirements of the tasks.

## 2nd GROUP OF TASKS (two hours)

Number 8-12

In this group of tasks you will

(1) examine the body structure of some water and terrestrial animals and

(2) estimate/determine some quantitative data (biomass and its losses) by the methods used in ecosystem research.

Start with tasks no. 10 and during free time (cooling of extract) you can do the remaining tasks (8, 9, 11 and 12) in any order.

Task No.8: Examination of the bodies of animals using Sheet P6, Table III, Table IV and Fig.1

Select 5 animals, belonging to three different systematic units. Three of your animals should be aquatic and two should be terrestrial, including the following: Organism "H" - an arthropod that moves on the surface of water, Organism "I" - an arthropod that can swim under the surface of water, a terrestrial animal that eats taller herbs (such as Rumex, Arctium, etc.) and a terrestrial arthropod that has a mouth that is labelled in Fig.1 by the capital letter C. To make your work in this Task easier, you may use:

- Fig.1: Types of Insect Mouthparts
- Table III: Anatomical Terms
- Table IV: Biosystematic Groups
- Provided examination tools

Read all of the following before continuing.

A. Observe the legs, mouths and wings of your animals.

B. Complete Sheet P6: "Ordering of the Animals Into Biosystematic Groups", using the number codes from Table IV as your answers.

C. Draw the details of the legs of animals "I" and "K" (their legs are specialized for certain ecological functions).

D. Make a schematic drawing of the wings of animals "J" and "K". Label them using the number codes from Table III.

E. Draw details of the legs of the animals I and K. Their legs are specialized for certain ecological functions.

F. Now put your animals into the correct vessels: H, I, J, K and Q).

Task No. 9: Analysis of a flower from plant E
Sheet P7, Table II, Score:

There is a flowering shoot of the modeled terrestrial plant E in a vessel. Make the following flower analysis, using the number codes from Table II as labels

a/ Attach single distinguished parts of the flower to the sheet P7 by a drop of gum/glue and identify them.
b/ Draw the important parts of the flower on the sheet.
c/ Write down the flower formula for analyzing flowers.
d/ Draw a pistil and/or the development of a fruit.

Task No. 10: Estimation of phytoplankton biomass in a fishpond
Sheet P8, Table V, Fig. 2, Score:

The biomass of phytoplankton in water is quantitatively estimated by several methods. The chlorophyll method, which you will be doing in very short version of, is one of them.
You will see the full full procedure in a videorecording. This method is also represented in Fig. 2.

In your Petri dish you will find a filter paper with phytoplankton on it. It was obtained by filtering 2 litres of water using the procedures drawn in Fig. 2.

a/ Cut the filter paper disc with the algae into small strips (app. 1x1 cm).
b/ Insert the cut strips into test tube with 10 ml of ethanol.
c/ Place the Flask into water bath and warm it to the ethanol~s boiling point (74 °C).
d/ Cool the mixture in the dark site (in refrigerator) for app. 30 minutes.
e/ Filter the cooled sample through a filter paper in a glass funnel into a volumetric Flask and add ethanol up to the line (use meniscus).
f/ Measure the extract~s level of transmission at wave lenght (according to the chlorophyll types) by spectrophotometer SPEKOL 11 (see Fig. 2) and at 750 nm (correction) Ask technical assistance for help. Obtain data and write it down on the Sheet P8.
g/ Calculate the concentrations of the chlorophyll A and in the extract - see the procedure in the Sheet P8. From the three different formula you must choose the correct one(s).
h/ Calculate the chlorophyll content in 1 m³ of water.

Task No. 11: Examination of the decomposition of plant matter from the deciduous forest (litter).
Sheet P8, Grid measure, Score:

In two Petri dishes you have soil samples from a deciduous forest stand. Three strips of filter paper (1x5 cm) were placed on each of them, in dish A, 40 days before and in the dish B, 14 days before.

a/ Examine the differences between dish A and dish B.

b/ Estimate the decomposion degree of the cellulose in per cent. Use the grid measure preparred for it. Write down the results in the sheet P8.

Task No. 12: Estimation of a biomass of trees in a forest stand.
Sheet P9, Table IV, Graph 1-3, Score:

There is a cross-section of a trunk of average trees from the stand on your table. The sample was cut from a height of 1.3 m above the soil surface. Height of the tree was 20 m.

a/ Estimate the age of the tree if you know that it was growing up to the height 1.3 m during 9 years.

b/ Measure the trunk thickness (including bark) at the height of 1.3 m. The thickness is designated as $d_{1.3}$ in grafs 1-3.

c/ Using the data you known, estimate, with the help of the graphs 1-3
- fresh matter of wood of the trunk (Graph 1),
- fresh matter of leaves (Graph 2)
- fresh matter of the total biomass (wood+bark+leaves) of the analysed mean tree (Graph 3).

Procedures of the biomass determination draw directly into the figures (Graphs 1-3). Write down your numerical data on the Sheet P9.

If you finished the second group of tasks, you may go over your answers before the answer sheets P6-P9 are collected.

# Tasks of IIIrd. International Biology Olympiad

Practical part

IIIRD GROUP OF TASKS (app. 60 min.)
Tasks No. 13 - 16

(Final Examination)

Dear Competitors,

In the previous two groups of tasks you have worked with organisms living in two ecological systems - in a water reservoir (fish pond) and in a deciduous forest stand.

In this final part you will analyze ecological relationships among observed/examined and other organisms. Look at the simplified scheme of the ecosystems in Fig. 3 and Fig. 4. You can find the organisms observed in the tasks No. 1-12. They are labelled by capital letters of the Latin alphabet. But you can also find other (new) organisms.

TASK NO. 13: Trophic Relationships in a Model of a Fresh Water Ecosystem
Sheet 10, Fig. 3

SCORE: ______

In Fig. 3 there are some organisms living in a fishpod. You can easily identify most of them. Plant C is a submersed plant, e.g. Elodea. Animal S is a representative of annelides (Annelida).

A. Identify the food interrelationships, considering food relationships (trophic levels), by using colored circles in the following way:

| | |
|---|---|
| Primary producers - green | Decomposers - blue |
| Primary consumers - violet | Terciary consumers - 2x red |
| Secondary consumers - red | $N_2$ Fixator - yellow |

B. Label by arrows, in the scheme of the ecosystem (Fig. 3), the food interrelationships (food web) of the organisms, e.g. label a predator - pre interrelationship with an arrow going from a predator to a prey.

C. Write down, on the back side of the Fig. 3 paper (only use one half), at least one completed food chain, which includes animal G. Use the capital letters in Fig. 3 for labeling of the individual organisms. Identify this food chain as TASK NO. 13 - PART C.

TASK NO. 13 (Continued):

D. Write down a food web that relates to animal G on the back side of the Fig. 3 paper (only use one half). Identify this food web as TASK NO. 13 PART - D.

TASK NO. 14: Trophic Relationships in a Modelled Forest Ecoystem
Sheet 11 (Fig.4), Fig.1, 3 and 5 Score:

In Fig.4 there are drawings of some organisms living in a forest ecosystem. In the left part of the drawing, you can find the following 3 organisms:

M - larva and adult (imago) with mouthparts marked in Fig.1 by letter "A".

L - larva with mouthparts of the type "A", but adult with type "B" (see Fig.1).

In the scheme of a forest ecosystem, open circles with arrows indicate the food of the animals or a living place.

A. Complete the open circles by using the capital letter of the organisms used in tasks No.1-12 or from Fig.3 and 4.

B. Mark by colored circles, considering trophic levels, the correct letters in the scheme (Fig. 4). Mark your answers as you did in Task 13, Fig.3.

C. Mark food chains, in the scheme (Fig.4), for organisms E, F, and I Include, in the food chains, some birds from Fig. 5 - according to their beak/bill. You can simply estimate the kind of food predominantly by them.

TASK NO.15: Comparison of Biomass of the Producers of a Fresh Water Ecosystem and a Forest Ecosystem (a very simplified example)
Sheet P12 (incl. Table V) Score:___

In tasks No.10 and 12 you have estimated/determined the biomass of the samples of selected ecological groups of organisms. Now you will compare the obtained data. Use 1 hectare (ha = 100m x 100m) as the comparative unit. 1 ha of a forest stand and 1 ha of a fishpond with a 1 m depth (wat column).

TASK NO. 15 (Continued)

A. Use the data in Sheet P8 to calculate the amount of the phytoplankton biomass in $1m^3$ and in 1ha of the fishpond. Assume that 1mg of chlorophyll is in $1m^3$ of water means 10g of fresh mass of algae.

B. Calculate, the amount of the total above-ground biomass of trees, using the data in the Sheet P9, in 1 ha of the forest stand (500 trees per 1 ha).

C. Calculate the total biomass of the primary producers in the forest, if you have assumed that biomass of herbaceous species was 10 ton (metric) per ha and the biomass of undergrowth organs (roots) of woody plants was 2ton per ha.

D. Compare the calculated data of the biomass of the producers in the modelled fishpond and in the modeled forest. Complete (write down) placing data into correct rectangle in competent ecological pyramids (pyramids of biomass) in the sheet P12.

E. Write down the number codes from Table V to the right of the correct trophic level of the pyramids.

TASK NO. 16: Decomposition of Litter in a Forest Stand. Sheet P13 SCORE:______

A. Claculate, using the data in sheet P9, the fresh matter of leaf biomass in all trees per 1ha of the stand.

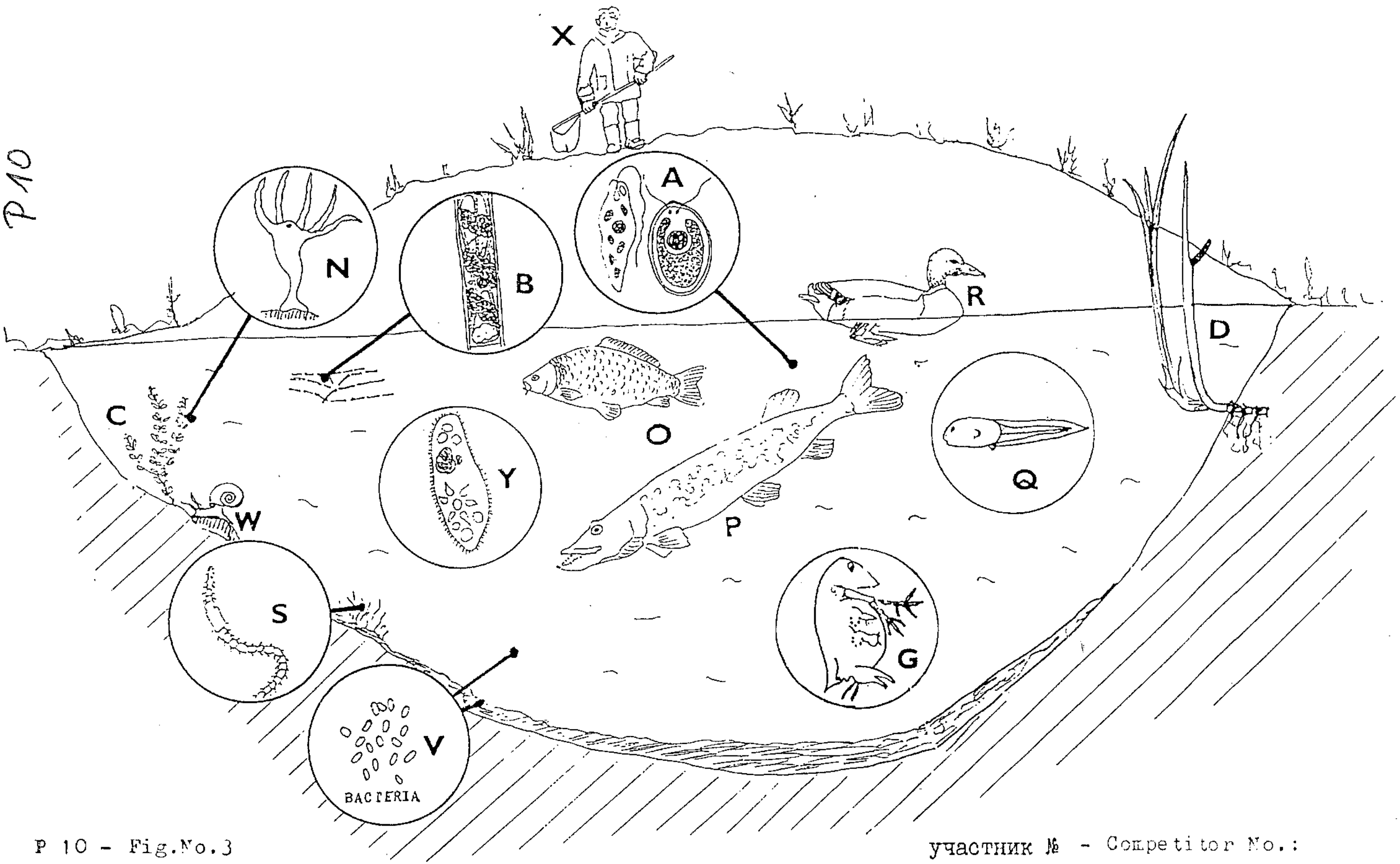
B. Calculate the amount of litter, if you know that the leaves fresh mass contains 70 per cent of water.

C. Calculate the time for the decomposition of the litter of the modelled forest ecosystem. Next follow these very simplified assumptions:

- the decomposition rate is the same as you determined in the Task No.11 (see sheet P9).
- all fallowed leaves were decomposed by the same decay rate (similar moisture conditions occurred)
- litter area is considered to be linearly related to the litter mass. it means that the unit area is decomposed by the some rate as mass unit.

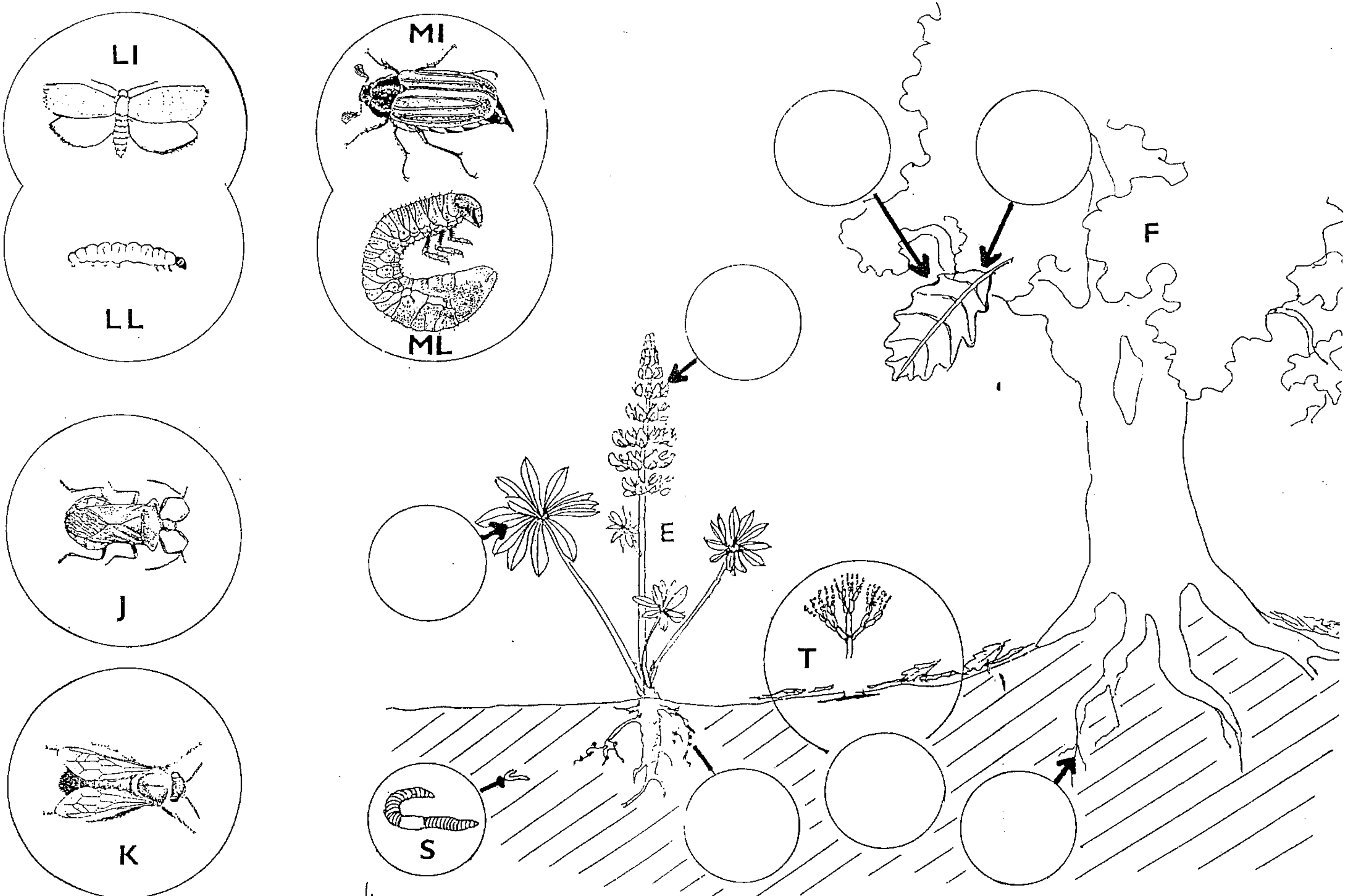
THE END!

P 10




P 10 - Fig.No.3

участник № - Competitor No.:




P 11 - Fig.No.4

УЧАСТНИК № - Competitor No.:

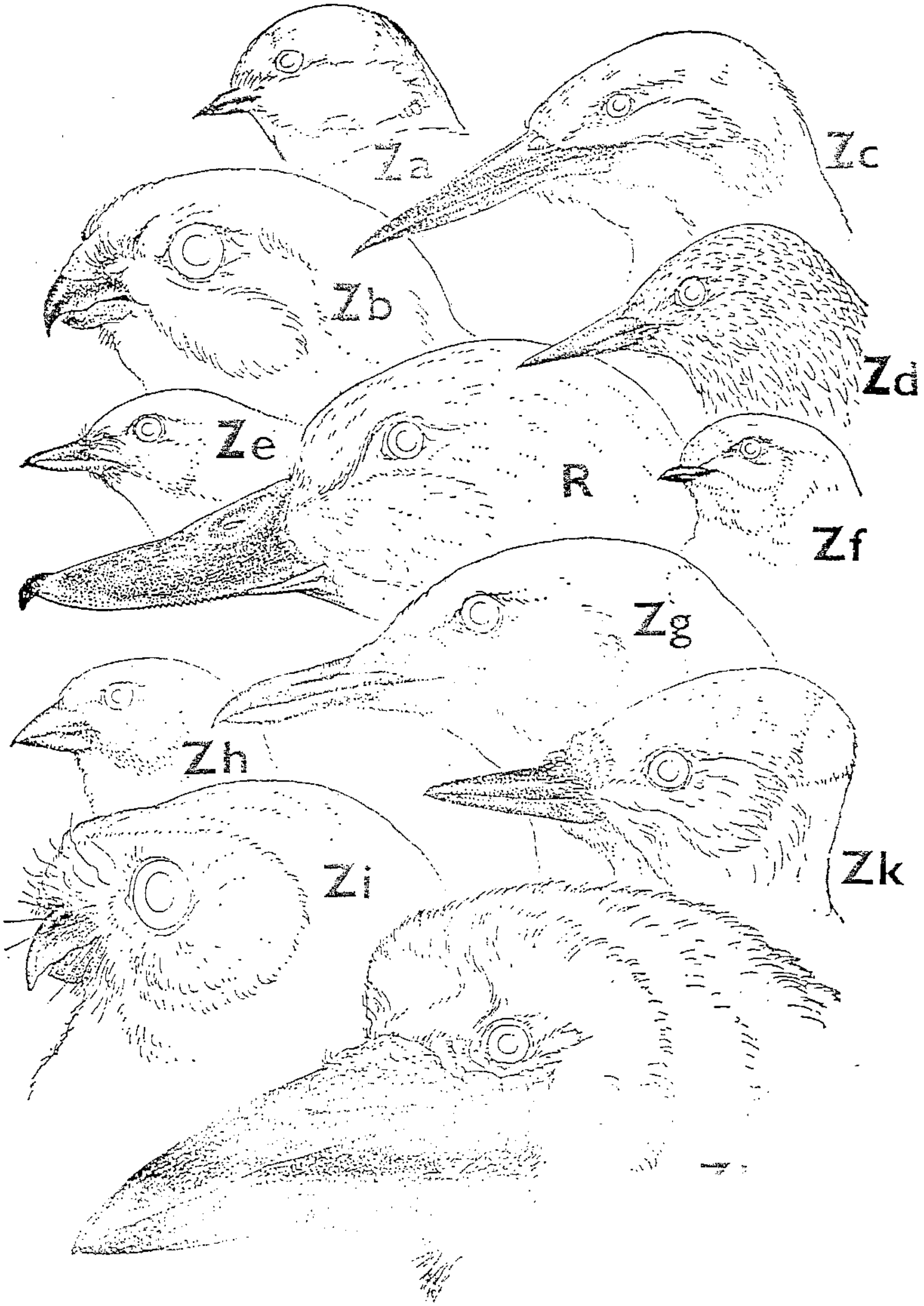



Fig. No. 5

Sheet P12 Competitor No. ..........

Vergleich der Biomasse in zwei Ökosystemen

Task 15 : Comparison of biomass of two ecosystems

Biomasse des Phytoplanktons

a/ Amount of phytoplankton biomass

- in 1 $m^3$ of water (Wasser) :
- in 1 ha of the fishpond (Teich) :

Biomasse der Bäume (oberirdisch)

b/ Aboue-ground biomass of trees:

- Biomasse eines mittleren Baumes
  biomass of average tree :
- Biomasse der Bäume im Wald
  biomass of trees in the forest:

Gesamtbiomasse der Primärproduzenten im Wald

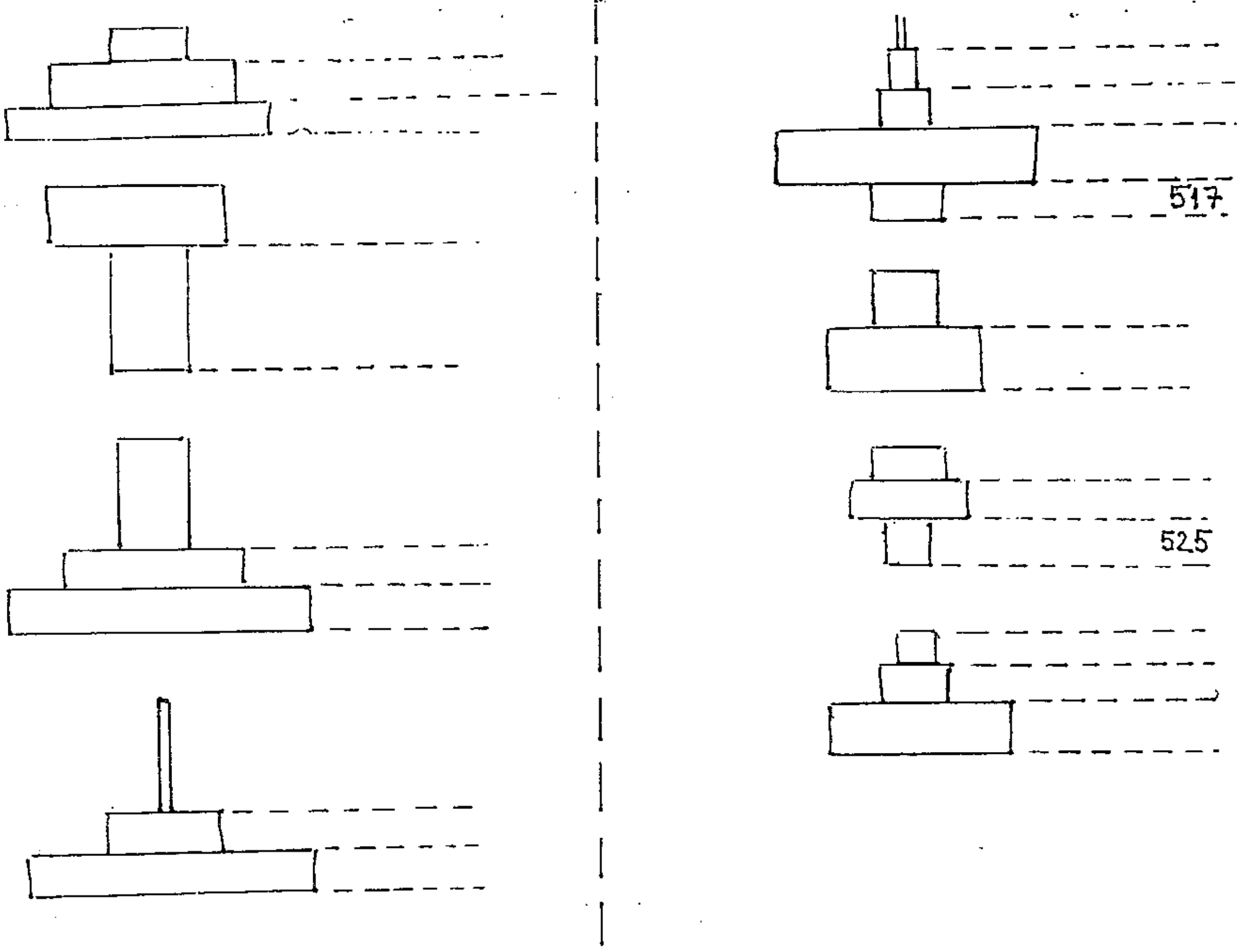
c/ Total biomass of primary producers in the forest

- woody above-gpound (oberirdische Teile) :
- woody roots Wurzeln : 2 000 kg
- herbs Kräuter : 10 000 kg
- Gesamtbiomasse
  total biomass :

Biomasse-Pyramiden

d/ Pyramids of biomass / some examples/:

| Pyramide<br>pyramids | Code der trophischen Ebenen<br>code of trofic level | pyramids | code of trofic leve |
|---|---|---|---|



P 13

SHEET 13

Frischgewicht der Biomasse der Blätter

A. Fresh matter of leaf biomass
(per 1 ha)

Gewicht des Abfalls im Wald

B. Amount of litter in the forest
stand:

Zersetzung des abgestorbenen Materials unter Berücksichtigung der Zeit

C. Time of decomposition of
all litter:

Table 5 Terms for trophic levels and food relationships

| | | |
|---|---|---|
| plant | 510 | Pflanze |
| animal | 511 | Tier |
| fungi | 512 | Pilze |
| bacteriae | 513 | Bakterien |
| producer | 514 | Produzent |
| consumer) | 515 | Konsument |
| autotroph | 516 | autotroph |
| decomposer | 517 | Reduzent, Destruent |
| herbivore | 518 | Pflanzenfresser |
| herb | 519 | Kraut |
| heterotroph | 520 | heterotroph |
| omnivore | 521 | Allesfresser |
| parasite | 522 | Parasit |
| pathogen | 523 | krankheitserregend |
| predator | 524 | Räuber |
| primary producer | 525 | Primärproduzent |
| primary consumer | 526 | Primärkonsument |
| prey | 527 | Beute |
| secondary consumer | 528 | Sekundärkonsument |
| tertiary consumer | 529 | Tertiärkonsument |
| zooplankton | 530 | Zooplankton |
| carnivore | 531 | Fleischfresser, Raubtiere |
| primary carnivore | 532 | primärer -"- |
| secondary carnivore | 533 | sekundärer -"- |
| detritovore | 534 | Detritusfresser, |
| top carnivore | 535 | Endkonsument |
| chemosynthetic autothroph | 536 | chemosynthetisch autotroph |
| photosynthetic autothroph | 537 | photosynthetisch autotroph |